

Automated File Organizer - Python Source Code

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import os
import shutil

# Predefined categories and their extensions
CATEGORIES = {
    "Documents": [".pdf", ".docx", ".txt", ".pptx", ".xlsx"],
    "Images": [".jpg", ".jpeg", ".png"],
    "Videos": [".mp4", ".avi", ".mkv"],
    "Audio": [".mp3", ".wav"],
    "Archives": [".zip", ".rar"]
}

def get_category(extension):
    """Return the category based on the file extension."""
    for category, extensions in CATEGORIES.items():
        if extension.lower() in extensions:
            return category
    return "Others"

def organize_files(directory):
    """Organize files in the specified directory based on their extensions."""
    if not os.path.exists(directory):
        print(f"Error: The directory '{directory}' does not exist.")
        return

    if not os.path.isdir(directory):
        print(f"Error: '{directory}' is not a directory.")
        return

    print(f"Scanning directory: {directory}\n")

    items = os.listdir(directory)
    files = [item for item in items if os.path.isfile(os.path.join(directory, item))]

    if not files:
        print("No files found to organize in the root of the directory.")
        return

    moved_count = 0

    for file in files:
        file_path = os.path.join(directory, file)
        _, extension = os.path.splitext(file)
        category = get_category(extension)

        category_path = os.path.join(directory, category)
        if not os.path.exists(category_path):
            os.makedirs(category_path)

        destination_path = os.path.join(category_path, file)

        try:
            shutil.move(file_path, destination_path)
            moved_count += 1
            print(f"Moved: {file} -> {category}/")
        except Exception as e:
            print(f"Error moving '{file}': {e}")

    print(f"\nSuccess! Successfully organized {moved_count} files.")

if __name__ == "__main__":
    print("--- Automated File Organizer ---")
    target_dir = input("Enter the path of the directory to organize: ").strip()

    if target_dir.startswith('.') and target_dir.endswith('.'):
        target_dir = target_dir[1:-1]
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        target_dir = target_dir[1:-1]
    elif target_dir.startswith('"') and target_dir.endswith('"'):
        target_dir = target_dir[1:-1]

    if target_dir:
        organize_files(target_dir)
    else:
        print("No directory path provided. Exiting.")
```